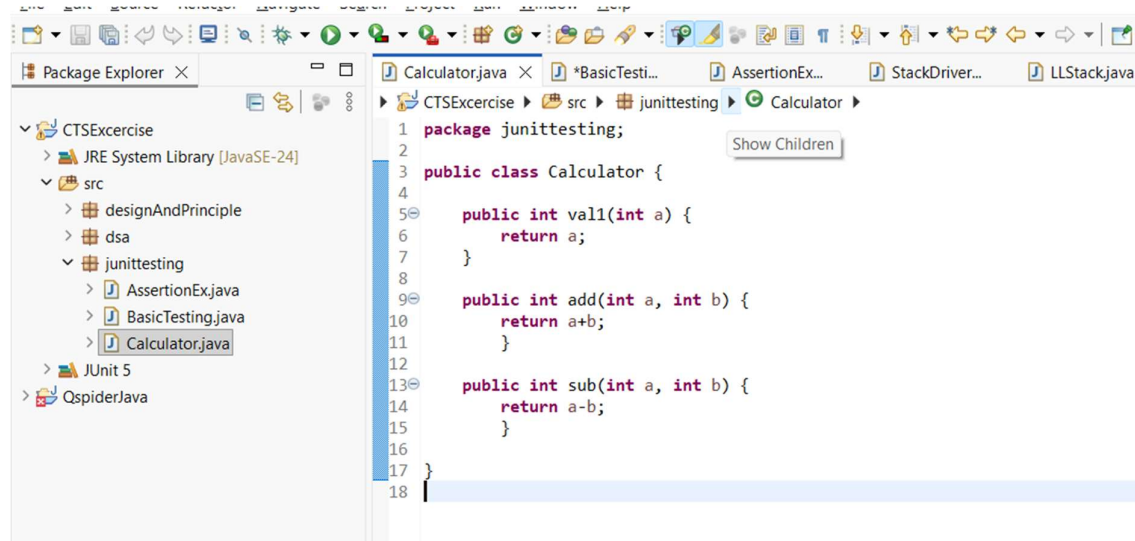
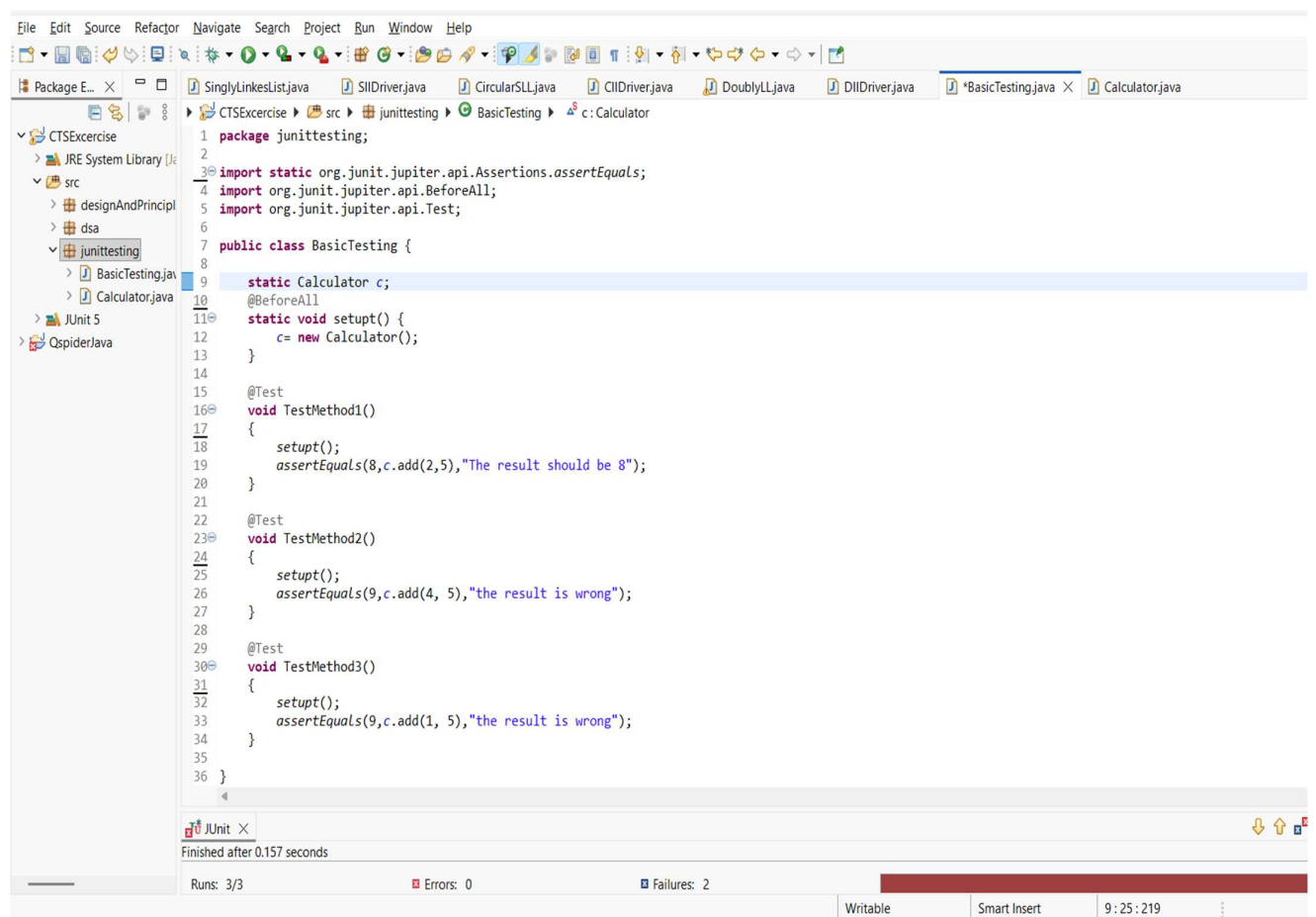


Exercise 1: Setting Up Junit and Exercise 2: Writing Basic JUnit Tests



The screenshot shows an IDE with the Package Explorer on the left and the Calculator.java file open in the editor. The Package Explorer shows the project structure: CTSExcercise > src > junittesting. The Calculator.java file contains the following code:

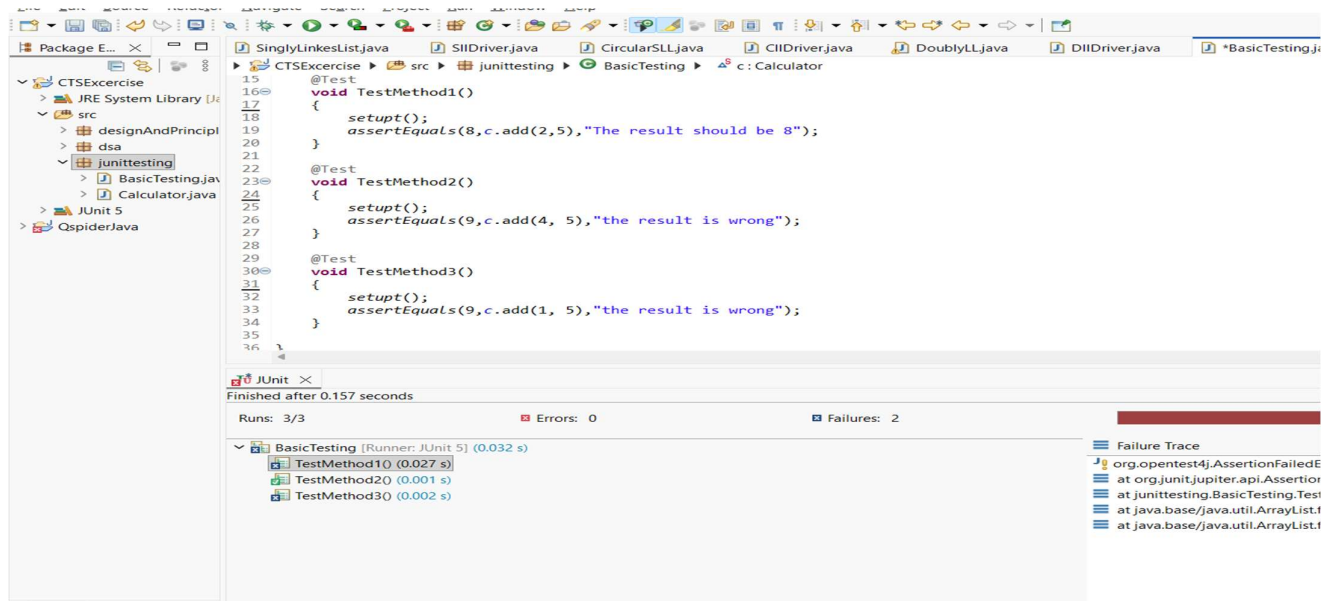
```
1 package junittesting;
2
3 public class Calculator {
4
5     public int val1(int a) {
6         return a;
7     }
8
9     public int add(int a, int b) {
10        return a+b;
11    }
12
13    public int sub(int a, int b) {
14        return a-b;
15    }
16
17 }
18
```



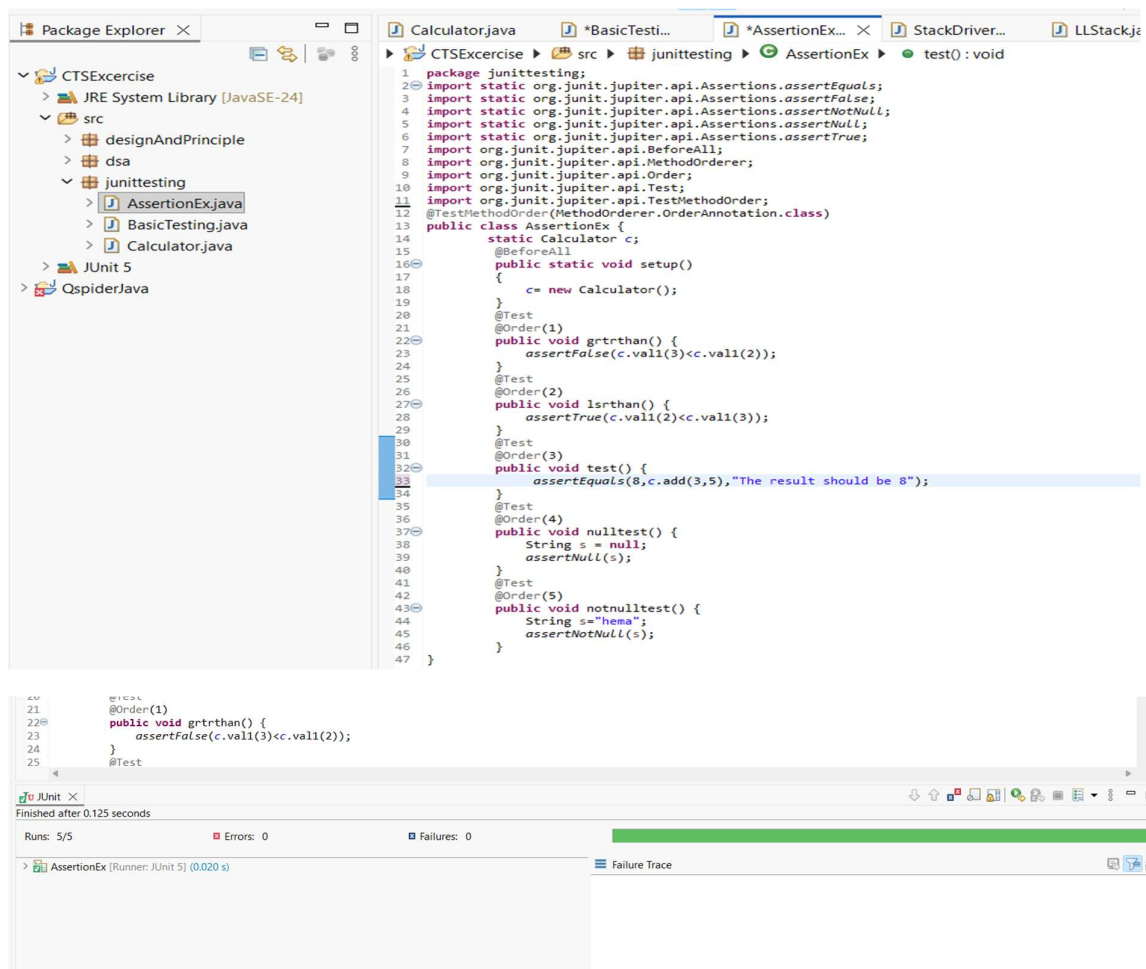
The screenshot shows the same IDE with the BasicTesting.java file open in the editor. The BasicTesting.java file contains the following code:

```
1 package junittesting;
2
3 import static org.junit.jupiter.api.Assertions.assertEquals;
4 import org.junit.jupiter.api.BeforeAll;
5 import org.junit.jupiter.api.Test;
6
7 public class BasicTesting {
8
9     static Calculator c;
10
11     @BeforeAll
12     static void setUp() {
13         c = new Calculator();
14     }
15
16     @Test
17     void TestMethod1() {
18         setUp();
19         assertEquals(8, c.add(2, 5), "The result should be 8");
20     }
21
22     @Test
23     void TestMethod2() {
24         setUp();
25         assertEquals(9, c.add(4, 5), "the result is wrong");
26     }
27
28     @Test
29     void TestMethod3() {
30         setUp();
31         assertEquals(9, c.add(1, 5), "the result is wrong");
32     }
33
34 }
35
36
```

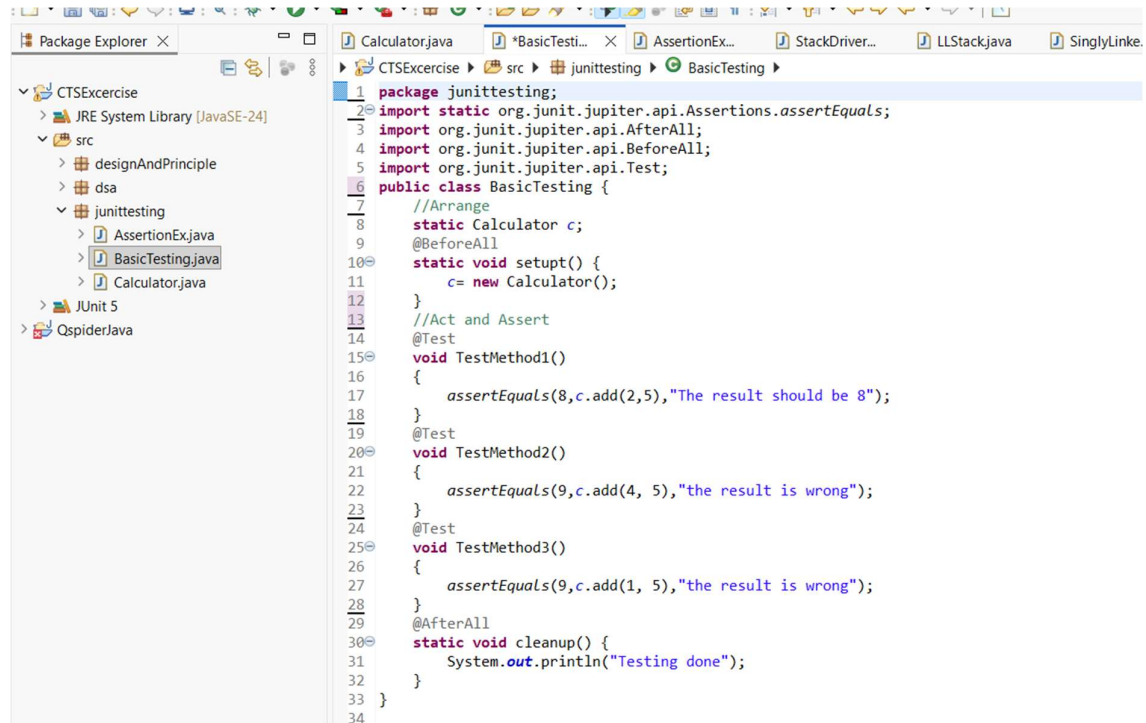
At the bottom of the IDE, a JUnit test run is shown. The status bar indicates: Runs: 3/3, Errors: 0, Failures: 2. The test run finished after 0.157 seconds. The bottom right corner shows the status bar with 'Writtable', 'Smart Insert', and the time '9:25:219'.



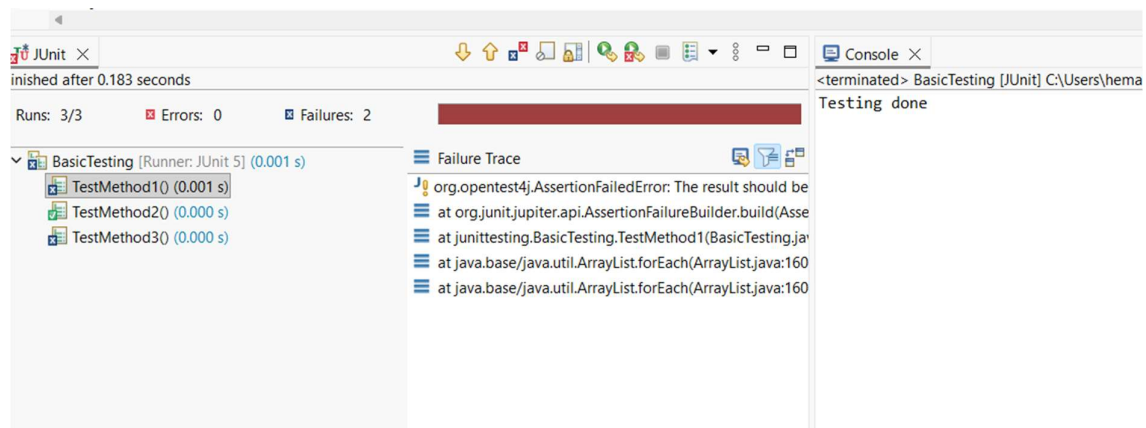
Exercise 3: Assertions in Junit



Exercise 4: Arrange-Act-Assert (AAA) Pattern, Test Fixtures, Setup and Teardown Methods in JUnit



```
1 package junittesting;
2 import static org.junit.jupiter.api.Assertions.assertEquals;
3 import org.junit.jupiter.api.AfterAll;
4 import org.junit.jupiter.api.BeforeAll;
5 import org.junit.jupiter.api.Test;
6 public class BasicTesting {
7     //Arrange
8     static Calculator c;
9     @BeforeAll
10    static void setup() {
11        c = new Calculator();
12    }
13    //Act and Assert
14    @Test
15    void TestMethod1()
16    {
17        assertEquals(8,c.add(2,5),"The result should be 8");
18    }
19    @Test
20    void TestMethod2()
21    {
22        assertEquals(9,c.add(4, 5),"the result is wrong");
23    }
24    @Test
25    void TestMethod3()
26    {
27        assertEquals(9,c.add(1, 5),"the result is wrong");
28    }
29    @AfterAll
30    static void cleanup() {
31        System.out.println("Testing done");
32    }
33 }
34
```



JUnit 5

inished after 0.183 seconds

Runs: 3/3 Errors: 0 Failures: 2

BasicTesting [Runner: JUnit 5] (0.001 s)

- TestMethod1() (0.001 s)
- TestMethod2() (0.000 s)
- TestMethod3() (0.000 s)

Failure Trace

```
org.opentest4j.AssertionFailedError: The result should be
at org.junit.jupiter.api.AssertionFailureBuilder.build(Asse
at junittesting.BasicTesting.TestMethod1(BasicTesting.java:
at java.base/java.util.ArrayList.forEach(ArrayList.java:160
at java.base/java.util.ArrayList.forEach(ArrayList.java:160
```

Console

```
<terminated> BasicTesting [JUnit] C:\Users\hema
Testing done
```